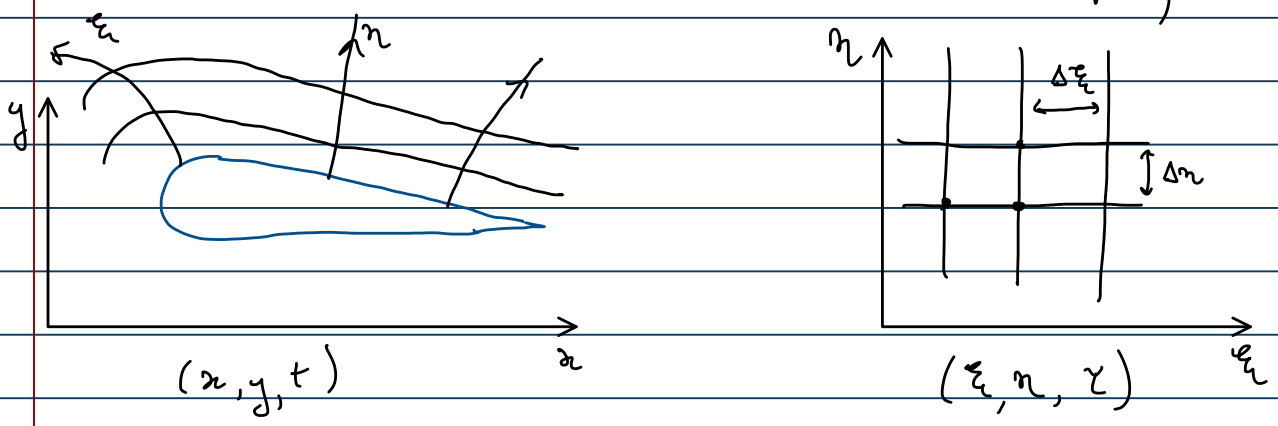
# Review 4.0

Saturday, 22 March 2025 7:14 PM

Grid generation

Physical grid ⟶ Computational grid (Uniform)




$$\xi = \xi(x, y, t)$$

$$\eta = \eta(x, y, t)$$

$$\tau = \tau(t)$$

$$\text{Jacobian matrix } (J) = \frac{\partial(x,y)}{\partial(\xi,\eta)} = \begin{vmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{vmatrix}$$